

development of advanced AI systems like autonomous machines.

- TensorFlow:** This is a FOSS library for data flow and differentiable programming across a range of tasks. It is a symbolic maths library, used for machine learning applications such as neural networks. It is used for both research and the development of various AI systems. Google has used TensorFlow for the development of its various products. Developed by the Google Brain team for internal use, it was released under the Apache License 2.0 on November 9, 2015. TensorFlow is available on 64-bit Linux, MacOS, Windows, and mobile computing platforms including Android and iOS. In May 2016, Google announced its Tensor Processing Unit (TPU), an application-specific integrated circuit (a hardware chip) built specifically for machine learning and tailored for TensorFlow. It's like an AI accelerator and Google has used TPUs in its data centres also. TensorFlow is an end-to-end open source platform for machine learning and is a rich system for managing all aspects of an ML system. TensorFlow APIs are arranged hierarchically. The high-level API is object-oriented and called `tf.keras`. It is another API variant of the FOSS tool Keras, and can be used to train machine learning models to make predictions. The TensorFlow toolkit also contains reusable libraries that can be used for learning algorithms. The low-level API of TensorFlow is used by many machine learning researchers to create and explore new ML algorithms. With the help of the toolkit, it is possible to create machine learning algorithms that can run on the CPU, GPU and TPU.
- TensorRT:** This is a free tool from Nvidia that comes as an SDK for the development of

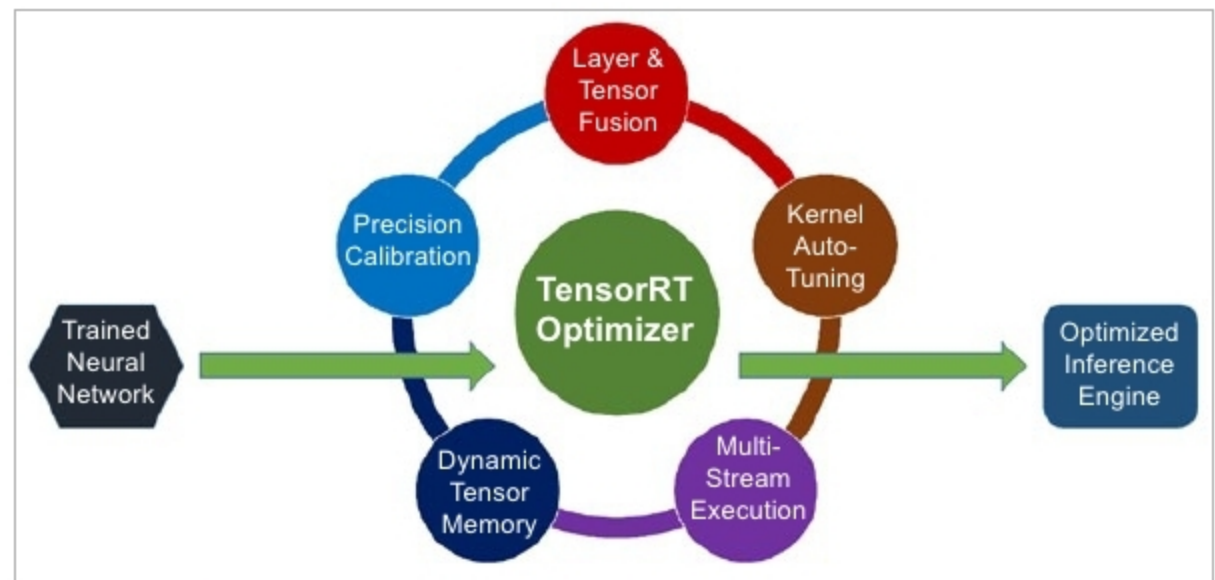


Figure 4: TensorRT optimiser

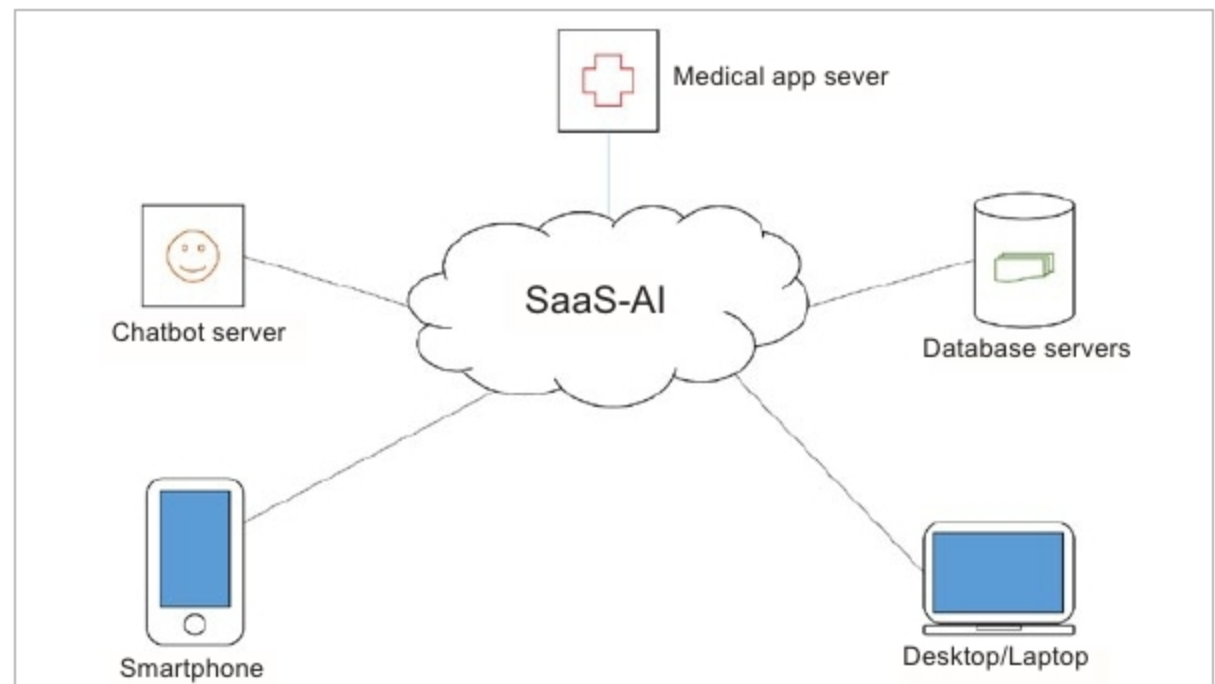


Figure 5: AI cloud medical assistance

high-performance deep learning inference. It includes a deep learning inference optimiser and runtime that delivers low latency and high-throughput for deep learning inference applications. The applications created with TensorRT work 40x faster than CPU-only platforms during inference. TensorRT can let you optimise neural network models that are trained in all major frameworks, calibrate them for lower precision with high accuracy and finally, you can deploy them on data centres, embedded systems or any automotive product. TensorRT is built on Nvidia's CUDA platform, which is an

API for developing AI systems and enables you to optimise inference for all deep learning frameworks — leveraging libraries, development tools and technologies in CUDA-X for artificial intelligence, autonomous machines, high-performance computing, and graphics. TensorRT provides INT8 and FP16 optimisations for production deployments of deep learning inference applications such as video streaming, speech recognition, recommendations and natural language processing. TensorRT has an optimiser that can do precision calibration, layer and tensor fusion, kernel auto-tuning, multi-stream execution and dynamic tensor memory handling.